

Book References: Coombes & Wedgwood – Chapter 4; Dayan & Abbott – Chapters 1, 5.8, 6; Gerstner et al – Chapters 3, 7 & 8

Code. Work in whatever language you prefer. Upload runnable code to the public GitHub repository named “CompNeuro2026”, labeled by problem, and give your account name and repository URL at the top of your submission. For coding problems, include the resulting labeled figures (axes, units, parameters) in your homework submission and report numerical results to 3–4 significant digits.

1. **Passive spread and active propagation on a cable.** A cable with an idealized (McKean-type) excitable nonlinearity obeys

$$\frac{\partial v}{\partial t} = D \frac{\partial^2 v}{\partial x^2} - v + H(v - a), \quad x \in \mathbb{R}, \quad (1)$$

where H is the Heaviside step and $0 < a < 1$ is the excitation threshold. Time is measured in membrane time constants, so the electrotonic length is $\lambda = \sqrt{D}$. This problem compares how far and how fast a signal gets with the nonlinearity switched off and switched on.

(a) First drop the H term, leaving the passive cable, and inject a unit charge at the origin at $t = 0$, so that $v(x, 0) = \delta(x)$. Show that writing $v = e^{-t}u$ turns the problem into the heat equation for u , and hence

$$v(x, t) = \frac{e^{-t}}{\sqrt{4\pi Dt}} \exp \left[-\frac{x^2}{4Dt} \right].$$

You can just show v solves the resulting equation by taking the appropriate partial derivatives v_t and v_{xx} and plugging them in. Show that the voltage at position x peaks at

$$t_{\text{peak}} = \frac{1}{4} \left[\sqrt{1 + \frac{4x^2}{D}} - 1 \right],$$

give its limits for $|x| \ll \lambda$ and $|x| \gg \lambda$, and show that the apparent propagation speed $|x|/t_{\text{peak}}$ tends to $2\sqrt{D}$ far from the injection site. Report t_{peak} at $x = 0.1, 1, 10$ for $D = 1$.

(b) Now restore the nonlinearity and look for a traveling front $v(x, t) = V(\xi)$, $\xi = x - ct$, with $V \rightarrow 1$ as $\xi \rightarrow -\infty$ and $V \rightarrow 0$ as $\xi \rightarrow +\infty$. Fix the translation by putting the threshold crossing at $\xi = 0$, so $V(0) = a$ and $V > a$ exactly for $\xi < 0$. Write the ODE for V , converting the t and x derivatives to ξ derivatives using the chain rule, solve it on each side of the crossing, and impose boundedness together with continuity of V and V' at $\xi = 0$ to derive

$$c = \sqrt{D} \frac{1 - 2a}{\sqrt{a(1 - a)}}.$$

(c) Read off the sign of c and determine the threshold at which the front stalls (where $c = 0$). Then compare c with the passive apparent speed from part (a) and find the value of a below which the active front is faster. Plot c against a over $0 < a < 1$ at $D = 1$, marking the passive apparent speed and the crossing. Finally, given that the diffusivity of a cable of diameter d scales as $D \propto d$, deduce how c scales with d , and say qualitatively what myelination changes, given that internode spacing grows in proportion to d .

2. **Conductance-based synapses, shunting, and the NMDA nonlinearity.** A passive compartment with excitatory and inhibitory synaptic conductances obeys

$$C \frac{dV}{dt} = -g_L (V - E_L) - g_e (V - E_e) - g_i (V - E_i),$$

with $g_e, g_i \geq 0$ held constant for this problem.

(a) Show the voltage relaxes exponentially to a conductance-weighted average of the three reversal potentials, and identify the effective time constant. Take $C = 200$ pF, $g_L = 10$ nS at rest, and $g_e + g_i = 40$ nS during bombardment, and report both time constants. What does synaptic bombardment do to the cell's response to a fixed injected current I ?

(b) Now take $E_i = E_L$, so the inhibitory synapse is purely shunting. Show that inhibition alone produces no steady voltage change at all, but that it still reduces the excitatory response, and that it does so divisively. Contrast this with a current-based inhibitory input, which injects a fixed negative current. Evaluate the gain reduction for $g_L = 10$ nS and $g_i = 20$ nS in the small g_e limit.

(c) The NMDA receptor conductance is voltage dependent through magnesium block,

$$I_N(V) = \bar{g} B(V) (V - E_N), \quad B(V) = \left[1 + \frac{[\text{Mg}^{2+}]}{3.57} e^{-V/V_0} \right]^{-1}, \quad V_0 = 16.13 \text{ mV},$$

with $[\text{Mg}^{2+}]$ in mM and $E_N = 0$ mV. Show that $B' = B(1 - B)/V_0$, and hence that

$$\frac{dI_N}{dV} = \bar{g} B \left[1 + \frac{(V - E_N)(1 - B)}{V_0} \right].$$

Find numerically, to three digits, the voltage at which the slope vanishes for $[\text{Mg}^{2+}] = 1$ mM, plot $I_N(V)$ over $V \in [-90, 0]$ mV for $[\text{Mg}^{2+}] = 0, 0.1, 1$ mM, and explain what the sign of the slope below that voltage means for the synapse.

3. **Inferring a Poisson rate.** A neuron is modeled as a homogeneous Poisson process of unknown rate ν , and we record K spikes in a total time T . Rather than the flat improper prior used in class, take a gamma prior

$$p_0(\nu) = \frac{b^a}{\Gamma(a)} \nu^{a-1} e^{-b\nu}, \quad a > 0, \quad b > 0,$$

which has mean a/b and variance a/b^2 .

(a) Show that the posterior is again a gamma density and identify its two parameters. Give the maximum likelihood estimate, the posterior mode, the posterior mean, and the posterior standard deviation.

(b) Show that the posterior mean is a weighted average of the prior mean and the MLE, identify the weights, and give the resulting interpretation of a and b as pseudo-data. Take a prior with mean 5 Hz and $b = 2$ s, record 30 spikes in 3 s, and report the MLE, the posterior mean, and the posterior standard deviation. How much recording time would it take to halve that standard deviation?

4. **Shot noise, balance, and the diffusion approximation.** A subthreshold neuron bombarded by synaptic events obeys

$$\tau \frac{dV}{dt} = -V + \tau \sum_k w \delta(t - t_k),$$

with V measured from rest, so each event steps the voltage instantly by w . Take the t_k to be a Poisson process of rate ν and ignore the threshold.

(a) Solve for $V(t)$ as a sum over past events. Then, by dividing the past into bins of width ds in which an event occurs independently with probability νds , show that in the stationary state

$$\langle V \rangle = \nu w \tau, \quad \text{Var}(V) = \frac{\nu w^2 \tau}{2}.$$

(b) Now let excitation of rate ν_E deliver steps $+w$ and inhibition of rate ν_I deliver steps $-w$. Write $\langle V \rangle$ and $\text{Var}(V)$, and note that the mean can be held fixed while the variance is made arbitrarily large. For $\tau = 20$ ms and $w = 0.5$ mV, find the rates giving $\langle V \rangle = 10$ mV and standard deviation 5 mV, and comment on what those numbers imply about the number of active synapses.

(c) The same binning argument gives every cumulant, not just the first two, since the n th cumulant of a sum of independent rare events is $\kappa_n = \nu \int_0^\infty [we^{-s/\tau}]^n ds$. Evaluate κ_n , show the skewness is

$$\frac{\kappa_3}{\kappa_2^{3/2}} = \frac{2\sqrt{2}}{3\sqrt{\nu\tau}},$$

independent of w , and state the precise sense in which the diffusion approximation, replacing the synaptic input by a Gaussian white noise with the same first two cumulants, becomes exact. Evaluate the skewness for the balanced numbers of part (b).

(d) Simulate the shot noise directly, holding $\langle V \rangle = 10$ mV and $\tau = 20$ ms fixed while taking $\nu\tau = 2, 20, 200$ with w set accordingly. Report the measured mean, variance, and skewness against theory in a table. For the two extreme cases $\nu\tau = 2$ and 200 only, overlay the voltage histogram with the matched Gaussian.

5. **Stationary density and firing rate of a noisy integrator.** Problem 4 replaced a synaptic barrage by a drift plus a white noise. Now put a threshold on such a model, dropping the leak so that the algebra stays elementary,

$$\frac{du}{dt} = \mu + \sqrt{2\sigma^2} \xi(t), \quad \mu > 0,$$

with ξ unit white noise, a spike emitted whenever u reaches $\theta > 0$, and reset to $u = 0$ immediately after. In the stationary state the density obeys $dJ/du = 0$ with probability current

$$J(u) = \mu p(u) - \sigma^2 \frac{dp}{du}.$$

(a) Argue that $J(u) = \nu$, the firing rate, for $0 < u < \theta$, while $J(u) = 0$ for $u < 0$. Each first order equation then has general solution

$$p(u) = \frac{\nu}{\mu} + Ce^{\mu u/\sigma^2} \quad (0 < u < \theta), \quad p(u) = Ae^{\mu u/\sigma^2} \quad (u < 0),$$

so you need only fix A and C . Use the absorbing condition $p(\theta) = 0$ and continuity of p at the reset to get

$$p(u) = \frac{\nu}{\mu} \begin{cases} 1 - e^{\mu(u-\theta)/\sigma^2}, & 0 \leq u \leq \theta, \\ [1 - e^{-\mu\theta/\sigma^2}] e^{\mu u/\sigma^2}, & u < 0. \end{cases}$$

Where does p fail to be differentiable, and by how much does the slope jump?

(b) Normalize p and show that

$$\nu = \frac{\mu}{\theta},$$

independent of σ . Reconcile this with the deterministic transit time from reset to threshold, and say what feature of this model is responsible, so that you can predict where the argument would break down. Then show the fraction of time spent below the reset is $\frac{\sigma^2}{\mu\theta} [1 - e^{-\mu\theta/\sigma^2}]$ and evaluate it for $\mu = \theta = 1$ at $\sigma^2 = 0.25, 1, 4$.

(c) Simulate the model by Euler-Maruyama with reset, taking $\mu = \theta = 1$ and $\sigma^2 = 0.25, 1, 4$. Histogram the subthreshold voltage and overlay $p(u)$ from part (a), and report the measured firing rate and the measured fraction of time below the reset against the predictions. Discard an initial transient before collecting statistics, and say why that matters here.